Collective dynamics of pedestrians with no fixed destination

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Abstract

Although many physical modeling approaches have been successful on reproducing variety of actual pedestrian phenomena in normal density, little is known about the behavior of highly dense crowd, either empirically nor theoretically. We here propose a simple self-propelled particle model that does not assume a fixed destination point. Instead, each particle is driven by a self-propelled force which aligns its orientation to the its own velocity vector, with a certain amount of delay. Although the particles interact with a short-ranged repulsive force, we find that a system consist of such particles exhibits a order-disorder transition as the delay time is altered. The result implies the conditions necessary for pedestrian crowd to spontaneously avoid the risk of excessive density, which sometimes leads to a tragic crowd disaster.

Keyword: pedestrian dynamics, self-propelled particles

1 Introduction

From schooling of fish, flocking of birds, swarming of insects to migration of cells or bacteria, we often observe collective behaviors of biological organisms.Recent studies have shown that these seemingly different phenomena could be treated under an unified concept: self-propelled particles[1].

Pedestrian dynamics is one of the subjects that has been discussed with such an idea during last two decades[2]. Most of the microscopic models assumes that each pedestrian drive themselves in the direction of their initially-given destinations. This assumption seems reasonable in normal situations, but observations shows that pedestrians may lose their way or abandon their initial destinations in an extremely dense crowd.

2 Model

We construct a simple self-propelled particle model which assumes no constant destination nor explicit alignment interaction. Let us consider a system of N polar disks of equal radius a moving in a 2D surface. The polarity of each disk is defined by $\hat{\mathbf{n}}_i = \cos\psi_i\hat{\mathbf{x}} + \sin\psi_i\hat{\mathbf{y}}$. The equation of motion is given by,

$$\frac{d\mathbf{v}_i}{dt} = f_0\hat{\mathbf{n}}_i - \gamma_v\mathbf{v}_i + \sum_j \mathbf{f}_{ij}. \qquad (1)$$

Particle i drives itself with a self-propulsion force of constant magnitude f_0 along its polarity axis $\hat{\mathbf{n}}_i$, while the velocity is damped by a drag force of coefficient γ_v. The dynamics of the polarity is overdamped, as

$$\frac{d\psi_i}{dt} = \gamma_\psi\left(\theta_i - \psi_i\right), \qquad (2)$$

where θ_i is the direction of the disk's velocity, i.e., $\mathbf{v}_i/v_i = \cos\theta_i\hat{\mathbf{x}} + \sin\theta_i\hat{\mathbf{y}}$. The only interaction between particles is linear-elastic pairwise repulsion.

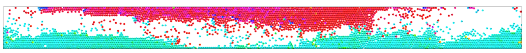


Figure1: Spontaneous formation of two bidirectiona lanes. The color indicates the orientation of the polarities. $N = 3200$, $\gamma_\psi = 0.01$.

3 Result

We performed numerical simulations altering the two parameters of the model, the angular damping parameter γ_ψ and the packing fraction ρ. The unit of length and time is chosen such that $2a = \gamma_v = 1$. The other parameters are fixed to be $f_0 = 1$ and $k = 100$. Starting from a state where orientation of polarities are randomly distributed, we investigated the evolution of the system.

We first carried out simulations with a "corridor" condition, i.e., in a rectangular area with periodic boundaries in the longitudinal direction while elastic repulsive boundaries in the width direction. The interaction between a particle and the repulsive boundary is assumed to be similar to the particle-particle interaction, that is, the interaction potential is elastic and frictionless. Interestingly, the simulations display bidirectional flows

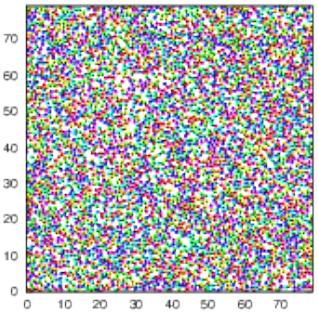


Figure2: Initial state, where the polarities are randomized. $N = 4000$.

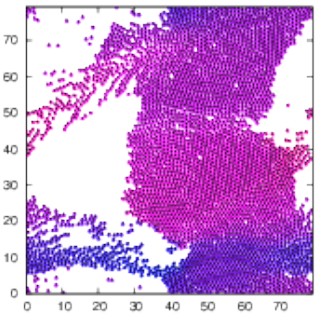


Figure3: Flocking state, where the motion of particles are aligned to each other. $N = 4000$, $\gamma_\psi = 0.1$.

separated into lanes (Fig. 1), which is a typical behavior for a pedestrian crowd[2].

For a system with periodic boundaries for both directions undergoes a transition from disordered gas-like state to an orientationally-ordered flocking state (as shown in Fig.3) to when γ_ψ is decreased. However, the growth of the order slows down and disordered state appears again for $\gamma_\psi \to 0$. Fig. 4 shows the time needed to develop order from initial disordered state.

This behavior is explained by assuming that the system is dilute enough so that only the uncorrelated, binary collisions take place, and both the velocities and the polarities are fully relaxed before each collision.

If the angular damping is weak, the polarities of two colliding particles remain unchanged, so the directions of motion are temporally changed by the collision but eventually restored to the original direction. Here the relative angle between the velocities does not change before and after the collision. By contrast, if the damping is strong, the polarities rotates themselves quickly to align to the directions of motion, so the particles moves as if they have exchanged their momentum. Here again the absolute value of the relative angle is maintained. For an intermediate damping parameter, the motion of two bodies align parallel due to the effect of the collision and the angular damping.

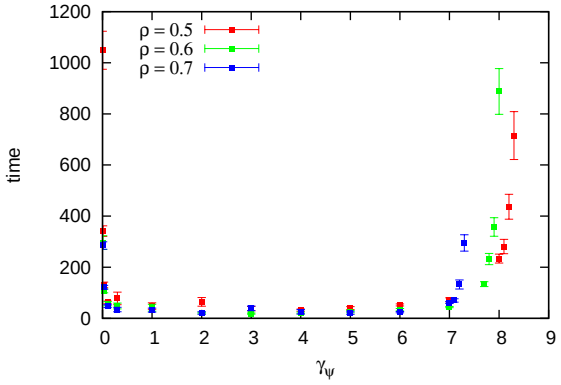


Figure4: The waiting time needed to develop spontaneous order.

Note that the collision process itself is elastic, i.e., no dissipation is taken into account. Nevertheless, the effective inelasticity introduced by the angular damping allows the formation of clusters, or "flocks," whose mechanism resembles to that underlies clustering phenomena of granular gas.[3].

4 Summary

Our model suggests that a long-range order can arise spontaneously by assuming that pedestrians interact solely with repulsive forces. Disordered phase appears in the domain in which the orientational damping is strong, suggesting that people are not capable of bailing out themselves from confusion under overcrowded conditions. We expect that the result can lead to a better understanding on the behavior of highly-dense crowd, which can cause a crowd disaster. We are looking forward to introducing more realistic interaction, effects of compression, falling, and panicking in mass accident in order to improve the model.

References

[1] T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Shochet, "Novel Type of Phase Transition in a System of Self-Driven Particles", Phys. Rev. Lett. 75, 1226 (1995).

[2] D. Helbing and P. Molnár, "Social force model for pedestrian dynamics", Phys. Rev. E 51, 4282 (1995).

[3] I. Goldhirsch and G. Zanetti, "Clustering instability in dissipative gases", Phys. Rev. Lett. 70, 1619 (1993).